1) A camera has a lens with a focal length of 5 cm. How far from the film must the lens be for a flower 0.7 m from the lens to be in focus? Describe the image.

Solution → Given ⇒ focal length = 35.0mm

object(flower) distance = 75.0mm.

Problem solving strategy

Step → To find position of image, find position of film.

⇒ distance between lens and film is image distance v_i.

⇒ We have given variable v_o and f.

⇒ Use lens formula

$$\frac{1}{v_o} + \frac{1}{v_i} = \frac{1}{f}$$

Now on rearranging and substituting value.

$$v_i = \left[\frac{1}{f} - \frac{1}{v_o}\right]^{-1} = \left[\frac{1}{35\times10^{-3}} - \frac{1}{75\times10^{-2}}\right]^{-1}$$

$$v_i = 3.67\text{cm}$$

So image distance will be 3.67 cm ans.

2) A 480-nm light goes through a 2-μm aperture. Find the first three angles of the dark spots.

3) Two monochromatic UV planewave light sources ($\lambda = 420\ nm$), are projected on a screen with 30° between them. Calculate the distance between any two bright spots on the screen.

4) Two monochromatic blue in-phase spherical light sources ($\lambda = 460\ nm$), 100 μm apart are projected on a screen 5 meters away. Calculate the location of the first three bright spots on the screen.

5) At visible light wavelengths, plastic has refractive index of 1.55. Calculate the percentage of power transmitted from air into plastic at normal incidence.

The percentage of power transmitted from air into plastic at normal incidence can be calculated using the following formula:

% Transmission = [(4n1n2)/((n1+n2)^2)] * 100

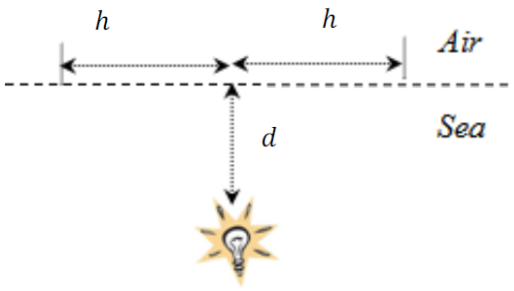
where n1 is the refractive index of the medium from which the light is incident (in this case, air), and n2 is the refractive index of the medium into which the light is transmitted (in this case, plastic).

Substituting the values given in the question, we get:

% Transmission = [(4 x 1 x 1.55)/(1 + 1.55)^2] * 100
% Transmission = 73.3%

Therefore, the percentage of power transmitted from air into plastic at normal incidence is 73.3%.

6) A light source that shines uniformly in all directions is submerged at a depth of d=2 meters below the surface of seawater (n=1.35). Assume that the sea surface is flat. Calculate the maximum value of h beyond which an observer at the sea-air interface will not see the light source.



7) Light that emerges out of two polarizers has 65% of the incident light intensity. Is the incident light linearly polarized, circularly polarized or unpolarized? Explain.

If the light intensity that emerges out of two polarizers is 65% of the incident light intensity, then we know that the two polarizers are not perfectly aligned with each other.

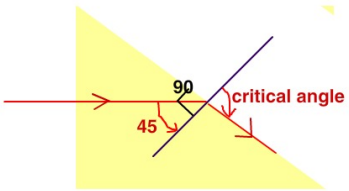
If the incident light was linearly polarized and the two polarizers were perfectly aligned, then the transmitted intensity would be 100%. However, since the transmitted intensity is only 65%, we can conclude that the incident light is not perfectly linearly polarized.

If the incident light was circularly polarized or unpolarized, then the transmitted intensity would be the same regardless of the orientation of the polarizers. Therefore, we can rule out circularly polarized light as the incident light since it is not affected by polarizers, and the transmitted intensity would be 100%. This leaves us with unpolarized light.

Unpolarized light is a mixture of all possible polarization states, and when it passes through a polarizer, it is reduced to half its intensity. If the transmitted intensity is 65%, this means that the two polarizers are not perfectly aligned, and some of the unpolarized light is able to pass through both polarizers. Therefore, we can conclude that the incident light is unpolarized.

8) A laser beam is incident perpendicular to the surface of one of the short faces of a 45–45–90 glass prism. If the refractive index of glass is 1.5, show that the light undergoes total internal reflection when it strikes the long face of the prism. Explain all steps with a sketch and label it.

(2.9) Total internal reflection occurs only when light passes from a more dense medium to a less dense medium at an angle of incidence greater than the critical angle.



critical angle for total internal reflection;
$\theta_c = 90^0$
Angle of incidence;
$\theta_i = 45^0$
for total internal reflection,
$n_1 sin\theta_i \geq n_2 sin\theta_c$

$$n_1 \geq \frac{n_2 sin\theta_c}{sin\theta_i}$$

$$n_1 \geq \frac{sin90}{sin45}$$

$$n_1 \geq 1.4142$$

Since, index of refraction for prism is 1.5 which is more than 1.4142, thus laser beam will be in total internal reflection.